Exp: 07

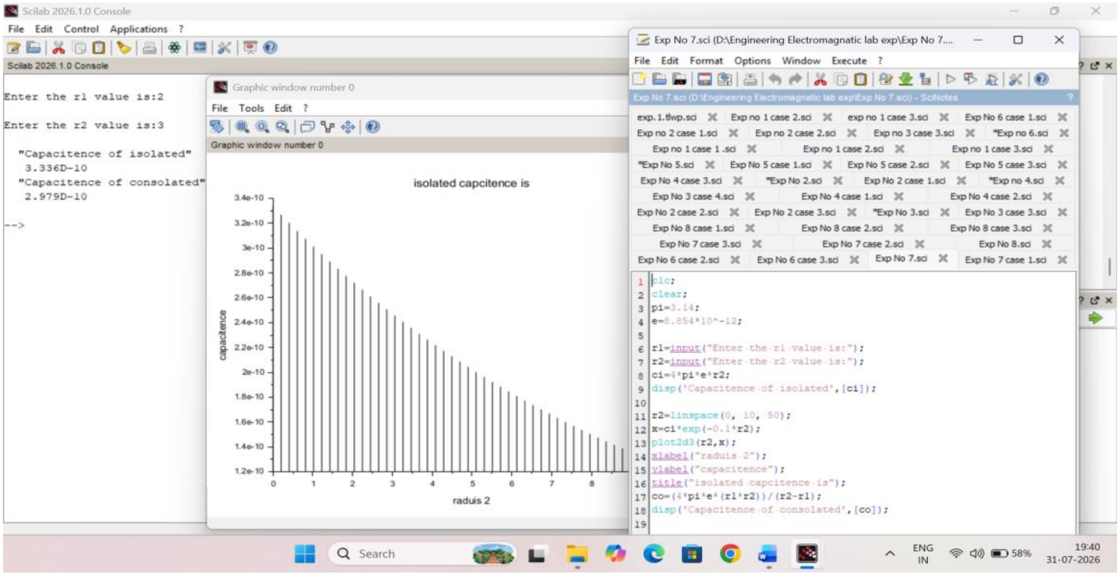
Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

Aim:

To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

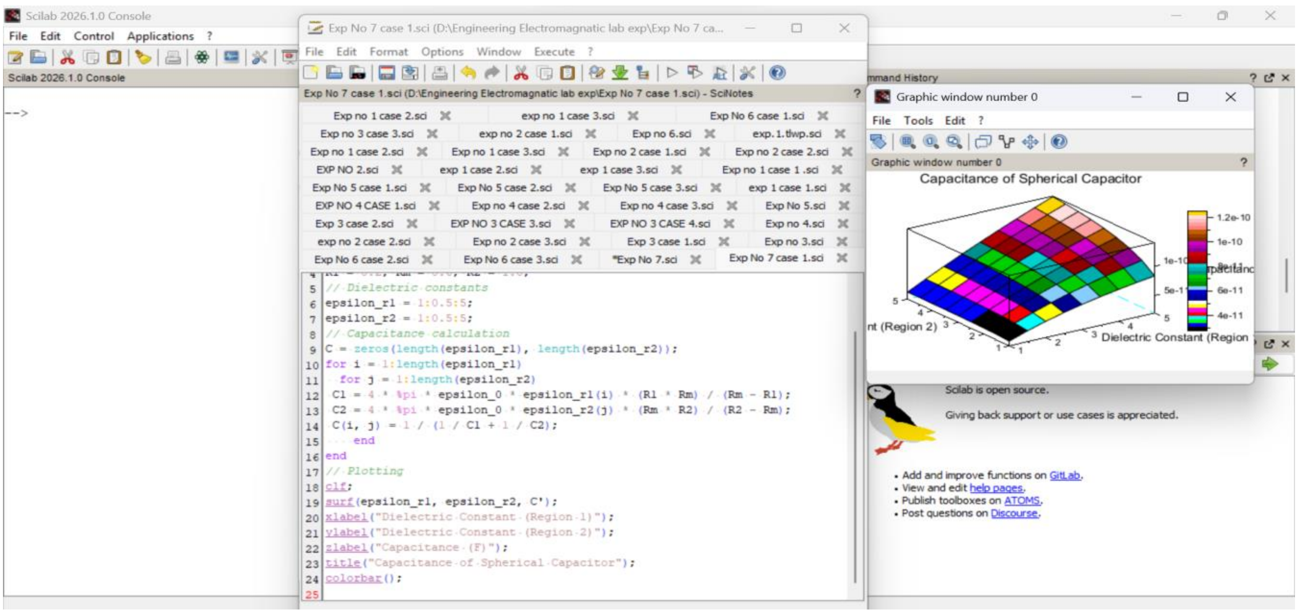
Software required:

•SCILAB version 6.0.1



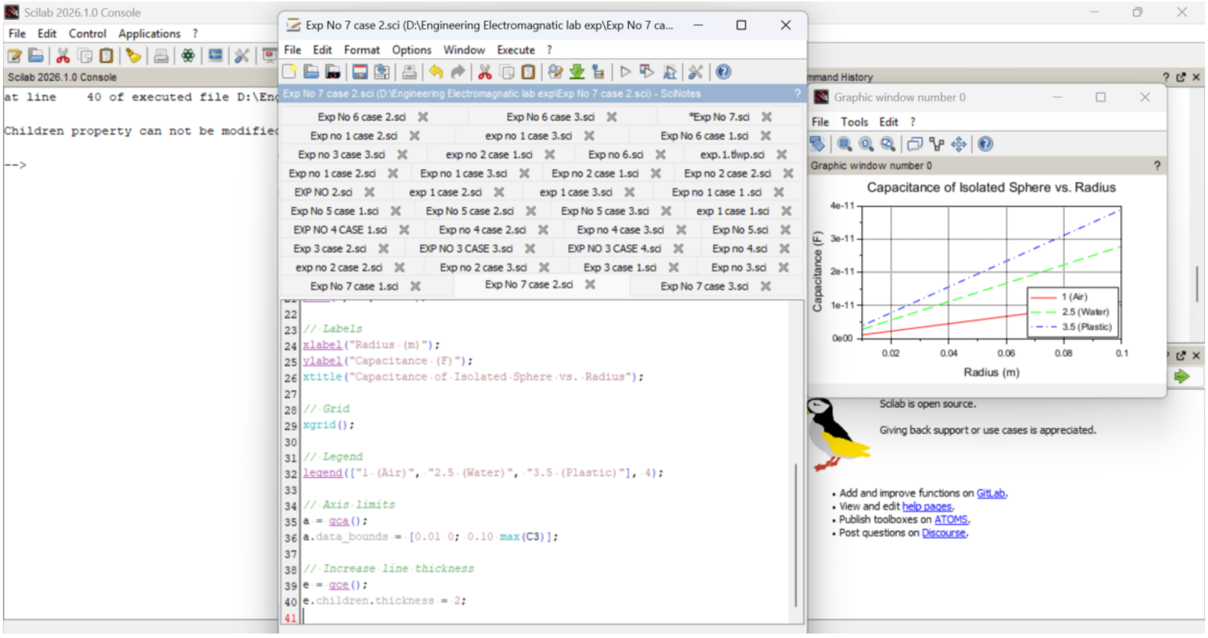
CASE 1:-CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS

CODE:-



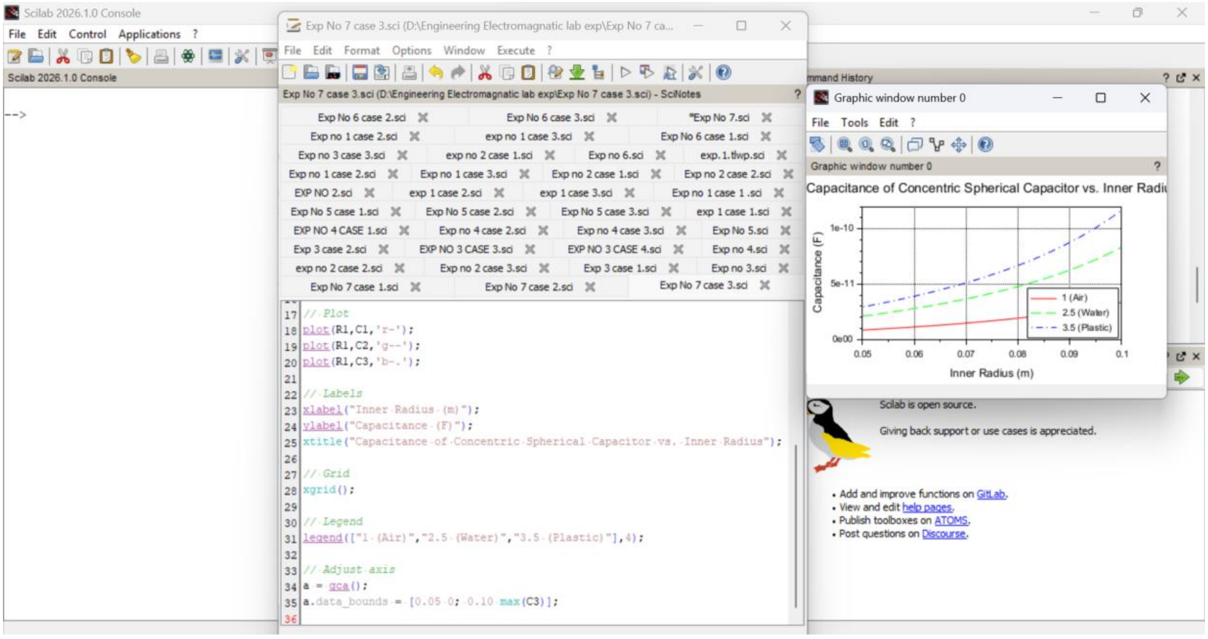
CASE 2: ISOLATED SPHERE

CODE:-



CASE 3: CONCENTRIC SPHERICAL CAPACITOR

CODE:-



Case 1: Concentric Spherical Capacitor with Different Dielectric Mediums

ϵ_{r1}	ϵ_{r2}	Capacitance (F)
1.0	1.0	2.73e-11
2.5	2.5	4.62e-11
5.0	5.0	7.12e-11

Case 2: Isolated Sphere

Radius (m)	Dielectric Constant (ϵ)	Capacitance (F)
0.01	1.0	1.11e-12
0.05	2.5	5.54e-12
0.10	3.5	12.33e-12

Case 3: Concentric Spherical Capacitor

Inner Radius (m)	Dielectric Constant (ϵ)	Capacitance (F)
0.05	1.0	1.13e-11
0.08	2.5	3.52e-11
0.10	3.5	6.11e-11

Result:

Thus, the program was verified for Capacitance of an isolated sphere and two concentric spheres using SCILAB was successfully.